ZOHAIB AHMED

AI / ML Engineer • Developer • CS Student

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PROFILE

AI/ML-focused developer and Computer Science undergraduate specializing in end-to-end machine learning and NLP applications — from data preprocessing and model training to evaluation and shipping models in interactive systems. Experienced with text classification, RAG and LLM integration, and comfortable across Python data science tooling and the MERN stack. Completed Coursera's Deep Learning Specialization, including transformer architectures.

TECHNICAL SKILLS

ML / Data Science	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, NLTK — classification, NLP (TF-IDF, sentiment analysis), preprocessing, feature engineering, model evaluation
Deep Learning	Neural networks, CNNs, Transformer architectures (Coursera Deep Learning Specialization)
AI / LLM Integration	RAG, Sentence Transformers, Ollama, Gemini API (function calling), Whisper (speech-to-text), TTS pipelines, prompt engineering
Languages	Python, Java, JavaScript
Backend	Node.js, Express.js, Flask, FastAPI
Frontend	HTML, CSS, JavaScript, React.js
Database	MongoDB (incl. Atlas), MySQL, SQL
Tools & Deployment	Git, GitHub, Streamlit, Vercel, JWT Auth, REST APIs

PROJECTS

Amazon Product Review Sentiment Analysis | *NLP + Machine Learning*
github.com/xizohaib/amazon-sentiment-analysis

- Built an end-to-end binary sentiment classifier on the Amazon Fine Food Reviews dataset (~568K reviews).
- Preprocessed text with NLTK, extracted TF-IDF features (20,000 features, unigrams + bigrams), and compared Logistic Regression vs. Multinomial Naive Bayes using accuracy, precision, recall, F1 and confusion matrices.

AI PDF Question-Answering System (RAG) | *Python, Streamlit, Ollama*
github.com/xizohaib/ai-pdf-qa-rag

- Built a PDF question-answering system with hybrid retrieval (TF-IDF + Sentence Transformers) merged via Reciprocal Rank Fusion, with optional cross-encoder reranking.
- Generates grounded answers with a local LLM (Ollama) with source pages, citation checks and abstention.
- Built a retrieval evaluation module: **93.3% Hit@5, 0.822 MRR and 0.049 s average retrieval time** on a labeled test set.

News Category Classification | *NLP + Streamlit App*
github.com/xizohaib/news-classifier-ml

- Developed an NLP pipeline that classifies news articles into categories: EDA, text cleaning, vectorization, model training and evaluation.
- Saved the trained model and served it in a Streamlit app that predicts categories for unseen articles.

Titanic Survival Prediction | *Machine Learning + Streamlit App*
github.com/xizohaib/Titanic-Survival-Prediction

- Trained a Random Forest classifier on the Kaggle Titanic dataset and deployed it as an interactive Streamlit web app with a custom nautical-themed UI and animated progress indicators for live predictions.

Loan Default Prediction System | *Machine Learning*
github.com/xizohaib/loan-default-prediction

- Built a classification model to predict loan defaults, covering data preprocessing, feature engineering, and model evaluation with scikit-learn.

AI-Based Interview Preparation System | *Full-Stack + AI*
github.com/xizohaib/Ai-based-Interview-Preparation-System

- Built a full-stack platform (React, Flask, MongoDB Atlas / MySQL) with authentication and a course dashboard for interview prep.
- Designed backend with an app-factory pattern, SQLAlchemy models, and JWT route protection; documented API with Postman.

Full-Stack Web Applications (MERN) | *Notes App, Todo App, Weather App*
github.com/xizohaib

- Full-stack MERN applications covering CRUD operations, REST API design, third-party weather API integration, and responsive UI.

CERTIFICATIONS

- Front-End Developer Certificate — Coursera / Google
- Google AI Fundamentals — Coursera / Google
- Deep Learning Specialization — Coursera / DeepLearning.AI

EDUCATION

Bachelor of Science in Computer Science
Sukkur IBA University • Expected Graduation: 2028